

Problem statement

Millions of speakers of African languages face barriers when using AI because most large language models are optimized for English and other high-resource languages. As a result, African languages are often fragmented into meaningless sub-word units, reducing model accuracy, increasing computational costs, and limiting access to reliable AI-powered services. Our project aims to address this challenge by developing a custom tokenizer that more accurately segments African languages into meaningful sub-word units using the SuperBPE framework, evaluated alongside Byte Pair Encoding (BPE) across representative languages from major African language families. Guided by the values of inclusivity, fairness, cultural preservation, transparency, and accountability, we seek to reduce linguistic bias, improve AI performance for African languages, and promote more equitable access to AI technologies for researchers, developers, and communities across Africa.